DATASET DETAILS

Online Retail II — Retail Transaction Analytics Platform

Dataset inventory, schema, preprocessing, quality findings, and known analytical considerations.

Project input file	Processed output
data/raw/Online_Retail_II.xlsx	data/processed/cleaned_retail.parquet

1. Dataset Identification

Item	Details
Dataset name	Online Retail II
Local source file	data/raw/Online_Retail_II.xlsx
Workbook sheets	Year 2009-2010; Year 2010-2011
Format	Excel workbook (.xlsx)
Project usage	Historical retail transaction analysis, sales summaries, product and customer analytics.
Public source	UCI Machine Learning Repository — Online Retail II dataset page
Source note	The project used the local workbook. Verify the downloaded file/version against the public source if exact provenance is required.

2. Dataset Purpose

The dataset records online retail transaction lines. It supports analysis of invoices, products, quantities, unit prices, transaction dates, customer identifiers, and countries. In this project, it feeds a repeatable ETL workflow and database-backed sales, customer, and product exploration.

3. Source Workbook Structure

Worksheet / measure	Profiling information
Year 2009-2010	525,461 records reported during initial profiling.
Year 2010-2011	Combined with the first worksheet. Exact individual row count was not retained in project notes.
Combined extraction	1,067,371 records reported by the ETL run.

4. Raw and Processed Schema

Field	Meaning / use
Invoice / invoice_no	Invoice identifier; normalized to invoice_no in processed schema.
StockCode / stock_code	Product or stock identifier.
Description / description	Product description text.
Quantity / quantity	Units on a transaction row; negative values may represent returns or adjustments.
InvoiceDate / invoice_date	Transaction date and time.
Price / unit_price	Unit price from the source workbook.
Customer ID / customer_id	Customer identifier; missing values occur in the source.
Country / country	Country associated with the transaction.
source_sheet	ETL-added field preserving the originating worksheet.
transaction_value	Derived row-level transaction value, based on ETL transformation logic.
is_cancelled_invoice	Derived flag for cancelled invoices.
is_negative_quantity	Derived flag for negative quantity rows.

The processed schema has 12 columns. Consult the project source code for exact transformation expressions and cancellation-identification rules.

5. Preprocessing Performed

The reported pipeline performs extraction, validation, transformation, cleaning, data-quality reporting, Parquet output, and MySQL loading. Field names are normalized and derived transaction/status flags are added. Duplicate rows and anomalous values are measured and handled according to the implemented cleaning logic.

Negative quantities may represent returns or corrections rather than invalid data. Missing customer IDs can leave transactions useful for sales totals while limiting customer-level analysis.

6. Data Quality Profile

Quality check	Reported count before cleaning
Total extracted records	1,067,371
Duplicate records	12,133
Negative quantity	22,950
Zero quantity	0
Negative price	5
Zero price	6,202
Invalid dates	0
Missing descriptions	4,382
Missing customer IDs	243,007

7. Processed Dataset and Database Results

Metric	Reported result
Cleaned dataset	1,055,233 rows × 12 columns
Processed file	data/processed/cleaned_retail.parquet
MySQL database / table	retail_analytics / transactions
Rows loaded to MySQL	1,055,233
Unique invoices	53,623
Unique products	5,131
Unique customers	5,942
Unique countries	43
Date range	2009-12-01 07:45:00 to 2011-12-09 12:50:00
Total quantity	10,575,682
Total transaction value	19,390,476.658

Currency is not explicitly asserted here because project notes do not establish a validated currency convention for all records.

8. Limitations and Interpretation Notes

• Record count falls from 1,067,371 to 1,055,233, a difference of 12,138. The quality report separately lists 12,133 duplicate records; reconcile the full difference against the implemented cleaning code.
• Missing customer IDs affect customer-level metrics and may result in fewer identifiable customers than transaction rows.
• Negative quantities, cancellations, and zero-price rows may be legitimate business events; interpretation depends on transaction and cleaning rules.
• The exact row count for the second worksheet and exact source-file version were not retained in the project notes.
• Raw and processed datasets are local artifacts. Avoid committing large datasets to GitHub unless required and permitted; use a small non-sensitive sample if needed.

9. Reproducibility

From the project root, ensure the workbook exists at data/raw/Online_Retail_II.xlsx, activate the Python environment, install dependencies from requirements.txt, and run:

```
python run_etl.py
```

Expected development results: 1,067,371 extracted records and 1,055,233 cleaned rows × 12 columns. Counts can differ if the workbook or code changes.

10. References

UCI Machine Learning Repository, Online Retail II: https://archive.ics.uci.edu/dataset/502/online+retail+ii
Project README: repository root README.md. ETL source: run_etl.py and modules under src/etl/. Database verification: project database test and SQL checks.